

»» Chapter II ««

THE COMPLEX FIELD. LIMITS

PREFATORY REMARKS

It is a commonplace that $\mathbf{R}$ is not *algebraically closed*; that is, it is not true that every polynomial of positive degree with real coefficients takes the value 0 at some $x \in \mathbf{R}$. A most simple example is $x^2 + 1$. Since $x^2 \geq 0$ for x real, $x^2 + 1 \geq 1 > 0$ for x real and hence $x^2 + 1$ does not take the value 0 for any $x \in \mathbf{R}$. Complex domain algebra and analysis begin with the search for a satisfactory field to operate in. The search is conditioned by the imposed demands. We might, for example, seek a field containing $\mathbf{R}$ as a subfield and at the same time algebraically closed. We might ask for ostensibly less, say, a field $\mathbf{C}$ containing $\mathbf{R}$ as a subfield and an element i satisfying $i^2 + 1 = 0$ such that $\mathbf{C}$ is generated by $\mathbf{R} \cup \{i\}$. This demand is a reasonable one and the consequences of the choice are fruitful. In particular, we shall see in §2, Ch. III (“fundamental” theorem of algebra) that $\mathbf{C}$ is algebraically closed. In fact, $\mathbf{C}$ is “the” minimal algebraically closed extension of $\mathbf{R}$. If F is an algebraically closed field containing $\mathbf{R}$, then there exists $i \in F$ satisfying $i^2 + 1 = 0$ and the subfield of F generated by $\mathbf{R} \cup \{i\}$ fulfills the conditions imposed on $\mathbf{C}$ and, trivially, is contained in F . We shall see in the next section that the $\mathbf{C}$ are algebraically equivalent.†

In §2 the geometric language of the complex field is introduced. Limit questions, infinite series, and power series are considered in the remaining sections of the chapter. The power series is part of the stock-in-trade of

† Hence the minimality property holds up to an isomorphism.

complex function theory and will be encountered frequently throughout the book.

1. THE COMPLEX FIELD

We consider a field $\mathbf{C}$ satisfying: (1) $\mathbf{R}$ is a subfield; (2) there exists $i \in \mathbf{C}$ such that $i^2 + 1 = 0$; (3) $\mathbf{C}$ is generated by $\mathbf{R} \cup \{i\}$. There are several easy remarks to be made. Given $a, b \in \mathbf{R}$, then $a + bi = 0$ if and only if $a = b = 0$. The sufficiency is trivial. From $a + bi = 0$ we obtain

$$a^2 + b^2 = (a + bi)(a - bi) = 0,$$

whence $a = b = 0$. Suppose that $a + bi \neq 0$. Then from

$$(a + bi)\left(\frac{a - bi}{a^2 + b^2}\right) = 1,$$

we see that

$$(a + bi)^{-1} = \frac{a}{a^2 + b^2} + \frac{(-b)}{a^2 + b^2} i.$$

It is now concluded that the set $\{a + bi: a, b \in \mathbf{R}\}$ is a subfield of $\mathbf{C}$ since differences and quotients of elements of the form $a + bi$, $a, b \in \mathbf{R}$, are also elements of the same form. If $a \in \mathbf{R}$, $a = a + 0i$; $i = 0 + 1i$. Hence $\mathbf{C} = \{a + bi: a, b \in \mathbf{R}\}$. Given $\alpha \in \mathbf{C}$, there is a unique $(a, b) \in \mathbf{R} \times \mathbf{R}$ such that $\alpha = a + bi$. For if $a + bi = c + di$, $a, b, c, d \in \mathbf{R}$, then $(a - c) + (b - d)i = 0$, whence the assertion. The rules for addition and multiplication,

$$\begin{aligned} (a + bi) + (c + di) &= (a + c) + (b + d)i, \\ (a + bi)(c + di) &= (ac - bd) + (ad + bc)i, \end{aligned} \tag{1.1}$$

are obvious and are stated only for reference.

We term $\mathbf{C}$ the *complex field*.[†] It is to be observed that the $\mathbf{C}$ are algebraically equivalent in the sense of isomorphism. That is, if $\mathbf{K}$ also fulfills the imposed conditions, there exists a univalent map f of $\mathbf{C}$ onto $\mathbf{K}$ satisfying $f(x + y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$, $x, y \in \mathbf{C}$.[‡] To see this, let $j \in \mathbf{K}$ satisfy $j^2 + 1 = 0$. A map f is given by

$$\{(a + bi, a + bj): a, b \in \mathbf{R}\},$$

[†] Again, a traditional definite article.

[‡] A map f of $\mathbf{C}$ onto $\mathbf{K}$ fulfilling these conditions is termed an *isomorphism* of $\mathbf{C}$ onto $\mathbf{K}$. The definition is, of course, applicable to more general algebraic structures.

as is easily verified. Note that it is harmless here to use the same symbols “+” and “·” relative to both $\mathbf{C}$ and $\mathbf{K}$.

EXERCISES

- 1.1. Verify that f so specified does have the stated properties.
 1.2. Determine the isomorphisms f of $\mathbf{C}$ onto itself satisfying $f(x) = x$, $x \in \mathbf{R}$.

From an axiomatic point of view we are on as good ground here as in the study of $\mathbf{R}$. However, just as one constructs $\mathbf{R}$ from more primitive structures, so here we may establish the existence of a $\mathbf{C}$ constructively. The construction given below in Ex. 1.3 is frequently used as the starting point of the study of the complex field. It will be seen that the construction is motivated by the preceding axiomatic considerations.

EXERCISE

1.3. Let $\mathbf{K} = \mathbf{R} \times \mathbf{R}$; let $+$ map $((a, b), (c, d))$ onto $(a + c, b + d)$; let $\cdot$ map $((a, b), (c, d))$ onto $(ac - bd, ad + bc)$. Show that $(\mathbf{K}, +, \cdot)$ is a field, that its zero is $(0, 0)$, its identity is $(1, 0)$, and that $[(0, 1)]^2 + (1, 0) = (0, 0)$. Show further that $\mathbf{R}_1 = \{(a, 0) : a \in \mathbf{R}\}$ is a subfield that is a complete ordered field with positive part $= \{(a, 0) : a > 0\}$, and that $\mathbf{K}$ is generated by $\mathbf{R}_1 \cup \{(0, 1)\}$.

We have seen (Ex. 4.15, §4, Ch. I) that from an algebraic point of view $\mathbf{R}$ is fairly simple. The only isomorphism of $\mathbf{R}$ onto itself is the identity map. We remark without proof that the set of isomorphisms of $\mathbf{C}$ onto itself is huge—a curious fact not needed for the purposes of classical analysis, but one which does show us that $\mathbf{C}$ is not without complication from an algebraic point of view.†

Real and imaginary part, conjugate, absolute value. We introduce certain useful maps of $\mathbf{C}$ into $\mathbf{R}$ and $\mathbf{C}$. Here $a, b \in \mathbf{R}$. (1) Re , the map taking $a + bi$ into a . We term the image of $z \in \mathbf{C}$ the *real part* of z . (Notational remark: We write $\text{Re } z$ for $\text{Re}(z)$ when it is convenient.) (2) Im , the map taking $a + bi$ into b . We term the image of z the *imaginary part* of z . Here, $\text{Im } z$ for $\text{Im}(z)$. (3) $\bar{}$, the map taking $a + bi$ into $a - bi$. We term the image of z the *conjugate* of z and denote it $\bar{z}$. (4) $||$, the map taking $a + bi$ into $\sqrt{a^2 + b^2}$. We term the image of z the *absolute value* of z and denote it $|z|$. The elements of the form ai , $a \in \mathbf{R}$, are termed *pure imaginary*.

We want to learn a number of the basic rules for calculating with Re , Im , $\bar{}$, $||$. The more obvious are stated without proof.

† To establish the remark one turns to transcendental field extensions (cf. [133, 139]).

(1) Re and Im are linear over $\mathbf{R}$ and each maps $\mathbf{C}$ onto $\mathbf{R}$. Re is a projection of $\mathbf{C}$ onto $\mathbf{R}$. In other words, given $z, w \in \mathbf{C}$, $a \in \mathbf{R}$, we have:

$$\text{Re}(z + w) = \text{Re } z + \text{Re } w, \quad \text{Re}(a z) = a \text{Re } z;$$

$$\text{Im}(z + w) = \text{Im } z + \text{Im } w, \quad \text{Im}(a z) = a \text{Im } z;$$

$$\text{Re}(\mathbf{C}) = \text{Im}(\mathbf{C}) = \mathbf{R}; \quad \text{Re}(\text{Re } z) = \text{Re } z.$$

(2) The conjugate map is an isomorphism of $\mathbf{C}$ onto itself. We have: $\overline{z + w} = \bar{z} + \bar{w}$, $\overline{z\bar{w}} = (\bar{z})(\bar{\bar{w}})$. Further, $\bar{\bar{z}} = z$ if and only if $z \in \mathbf{R}$. The conjugate map is an *involution*, that is, $\bar{\bar{z}} = z$, $z \in \mathbf{C}$.

$$(3) \text{Re } z = (z + \bar{z})/2, \quad \text{Im } z = (z - \bar{z})/2i.$$

$$(4) |z|^2 = z\bar{z}.$$

The following observation is to be made: for elements of $\mathbf{R}$ the definition of absolute value is consistent with that given earlier. In fact, given $a \in \mathbf{R}$, $a = a + 0i$, so that the present definition of $|a|$ yields $\sqrt{a^2}$, and this is precisely the absolute value of a in the sense of real analysis. We recall the convention on $\sqrt{\quad}$ in $\mathbf{R}$.

Two important rules pertain to the absolute value of a sum and the absolute value of a product. They are

$$|z + w| \leq |z| + |w|, \tag{1.2}$$

and

$$|zw| = |z| |w|. \tag{1.3}$$

For (1.3) we may proceed very simply as follows:

$$|zw|^2 = (zw)(\overline{zw}) = |z|^2 |w|^2 = (|z| |w|)^2.$$

Hence $|zw| = |z| |w|$ since x^2 defines an increasing function on $\mathbf{R}^+$. For (1.2) we note that

$$\begin{aligned} |z + w|^2 &= |z|^2 + |w|^2 + 2 \text{Re}(z\bar{w}) \\ &\leq |z|^2 + |w|^2 + 2 |z\bar{w}| \\ &= (|z| + |w|)^2. \end{aligned}$$

Hence $|z + w| \leq |z| + |w|$. We note that equality holds if and only if $z\bar{w} \geq 0$.

EXERCISES

1.4. Let $U = \{u \in \mathbf{C}, |u| = 1\}$. Show that U is a subgroup of $\{\mathbf{C} - \{0\}, \cdot\}$.

1.5. $z\bar{w} \geq 0$ if and only if there exist $\lambda, \mu \geq 0$ but not both zero such that $\lambda z = \mu w$.

- 1.6. $|z| \leq |\operatorname{Re} z| + |\operatorname{Im} z| \leq 2^{1/2} |z|$. When does equality occur on either side?
- 1.7. $|\operatorname{Im}(z^k)| \leq k |\operatorname{Im} z| |z|^{k-1}$, k a positive integer.
- 1.8. $|z + w|^2 + |z - w|^2 = 2(|z|^2 + |w|^2)$.
- 1.9. Given $z_k \in \mathbb{C}$, $k = 1, \dots, n$. Show that $|\sum_1^n z_k| \leq \sum_1^n |z_k|$. Under what conditions does equality hold?
- 1.10. Show that if $z^{n+1} + \sum_0^n a_k z^k = 0$, $n \in \mathbb{N}$, $\sum_0^n |a_k| > 0$, then $|z|$ does not exceed the unique positive root of $x^{n+1} - \sum_0^n |a_k| x^k = 0$ which in turn is not greater than $\max\{\sum_0^n |a_k|, 1\}$ (cf. Ex. 4.14, Ch. I). Hint: From $z^{n+1} = -\sum_0^n a_k z^k$, conclude $|z|^{n+1} \leq \sum_0^n |a_k| |z|^k$.
- 1.11. Given a_k real, $k = 0, \dots, n+1$, satisfying $a_0 \geq a_1 \geq \dots \geq a_{n+1} > 0$. Show that if $\sum_0^{n+1} a_k z^k = 0$, then $|z| \geq 1$ (Eneström). Hint: Apply the Abel identity to $\sum_0^{n+1} a_k z^k$.

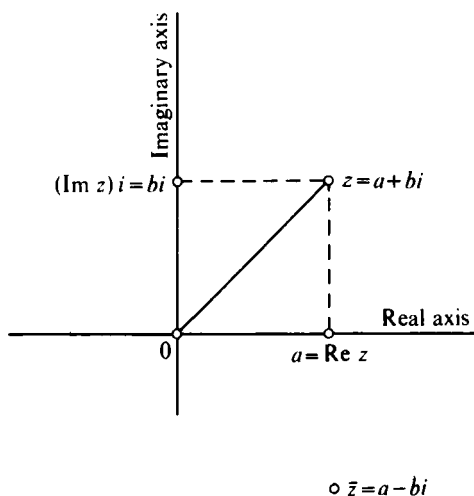


FIG. 2.1

2. GEOMETRY OF THE COMPLEX PLANE

The map $\sigma: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{C}$ taking (a, b) into $a + bi$ is a univalent map of $\mathbb{R} \times \mathbb{R}$ onto $\mathbb{C}$. We term σ the *canonical map* of $\mathbb{R} \times \mathbb{R}$ onto $\mathbb{C}$. It is to be noted that absolute value in $\mathbb{C}$ permits us to introduce a notion of distance in $\mathbb{C}$. That is, given $z, w \in \mathbb{C}$, we define the *distance* from z to w as $|w - z|$ and we see that distance so defined is a metric, for it satisfies $|w - z| = 0$ if and only if $w = z$, $|z - w| = |w - z|$, $|w - z| \leq |u - z| + |w - u|$.† Given

† The concept of a metric will be considered in Ch. III.

$(a, b), (c, d) \in \mathbf{R} \times \mathbf{R}$, we see that

$$|\sigma(c, d) - \sigma(a, b)| = \sqrt{[(c - a)^2 + (d - b)^2]} \quad (2.1)$$

The relation (2.1) asserts that σ preserves distance.

Using σ and the coordinatization of the Euclidean plane induced by the introduction of a rectangular coordinate frame we coordinatize the Euclidean plane via $\mathbf{C}$. We shall term such a coordinatization of the Euclidean plane *Gaussian* or complex. We note that $\mathbf{R}$ corresponds to the first axis of the frame which we term the *real axis* and the set $\{ai: a \in \mathbf{R}\}$ corresponds to the second axis, which we term the *imaginary axis*. The complex number 0 corresponds to the origin. $\text{Re } z$ is the coordinate of the orthogonal projection on the real axis of the point with complex coordinate z . Corresponding geometric interpretations for $(\text{Im } z)i$, $\bar{z}$, and $|z|$ are left to the reader.

While the preceding paragraph serves to relate $\mathbf{C}$ with 2-dimensional Euclidean geometry, for the purposes of complex function theory we may as well operate directly in $\mathbf{C}$. Thus we call $\mathbf{C}$ itself the *Gaussian plane* and introduce all the geometric concepts to be employed in terms of the identity map of $\mathbf{C}$ onto itself and the operations and special maps (Re , Im , $\bar{}$, $||$) already introduced. Thus $\mathbf{R}$ is the real axis and $\{ai: a \in \mathbf{R}\}$ is the imaginary axis. We are dealing with a realization of 2-dimensional Euclidean geometry. Starting this way, the reader need not concern himself with the refinements inherent in an axiomatic account of 2-dimensional Euclidean geometry. We shall see that thanks to the canonical map σ we obtain a "faithful" copy of the conventional 2-dimensional coordinate geometry.

We introduce several notions and notations that will be constantly employed. By a *straight line* is meant a set of the form $\{a + bt: t \in \mathbf{R}\}$, where $a, b \in \mathbf{C}$, $b \neq 0$. Given $a, b \in \mathbf{C}$, by the *segment* $[a, b]$ is meant the set $\{(1 - t)a + tb: 0 \leq t \leq 1\}$. It should be verified that, when $a, b \in \mathbf{R}$, $b \geq a$, the present definition is consistent with that of §3, Ch. I. Given $a, b \in \mathbf{C}$, $b \neq 0$, by the *ray* associated with (a, b) is meant the set $\{a + bt: t \in \mathbf{R}^+\}$.

EXERCISES

- 2.1. Given $a, b (\neq a) \in \mathbf{C}$. There exists a unique line containing a and b . In fact, it is $\{(1 - t)a + tb: t \in \mathbf{R}\}$. What does this say about segments?
- 2.2. A ray admits a unique representation of the form

$$\{a + ut: t \in \mathbf{R}^+\}, \quad a \in \mathbf{C}, |u| = 1.$$

- 2.3. Given $a, b \in \mathbf{C}$, $b \neq 0$. Let $L(a, b) = \{a + bt: t \in \mathbf{R}\}$. Determine a necessary and sufficient condition for $L(a_1, b_1) = L(a_2, b_2)$, for $L(a_1, b_1) \cap L(a_2, b_2) = \emptyset$, where the (a_k, b_k) are admitted (a, b) .

Given $a \in \mathbb{C}$, $r \in \mathbb{R}^+$, we denote by $C(a; r)$ the set $\{|z - a| = r\}$. It is the *circle* (or *circumference*) with center a , radius r . For positive r we denote by $\Delta(a; r)$ the set $\{|z - a| < r\}$. It is the *circular disk* (alias *open ball*) with center a , radius r .

EXERCISE

2.4. If $\Delta(a; r) = \Delta(b; s)$, then $a = b$ and $r = s$.

Convexity. We say that $E(\subset \mathbb{C})$ is *convex* provided that $[a, b] \subset E$, whenever $a, b \in E$. Convex sets have many interesting properties. We shall encounter the notion of convexity in the theorem of Gauss concerning the roots of the derivative of a polynomial (§9, Ch. IV). Given $A \subset \mathbb{C}$, there is a smallest (in the sense of set inclusion) convex set containing A . In fact, it is the intersection of all convex sets containing A . It is termed the *convex hull* of A .

EXERCISES

2.5. The following sets are convex: $\mathbb{C}$; $\Delta(a; r)$; $[a, b]$; $\{a < \operatorname{Re} z \leq b\}$, $a < b$; $\sigma([a, b] \times [c, d])$, $a < b$, $c < d$.

2.6. Given A convex; n a positive integer; $a_k \in A$, $\mu_k \geq 0$, $k = 1, \dots, n$; $\sum_1^n \mu_k = 1$. Then $\sum_1^n \mu_k a_k \in A$. Hint: Proof by induction.

2.7. Let A be a given subset of $\mathbb{C}$; let n be a positive integer. Let B_n denote the set of sums of the form $\sum_1^n \mu_k a_k$, where $a_k \in A$, $\mu_k \geq 0$, $k = 1, \dots, n$, and $\sum_1^n \mu_k = 1$. Show that the convex hull of A is $\bigcup_{k \in \mathbb{N}} B_{k+1}$. Show that if A is the finite set $\{a_1, \dots, a_n\}$, then the convex hull of A is B_n .

Linear maps of $\mathbb{C}$ into $\mathbb{C}$. Given $a, b \in \mathbb{C}$, we consider the linear function l with domain $\mathbb{C}$ given by

$$l(z) = az + b. \quad (2.2)$$

We put aside the trivial case where $a = 0$ and seek to interpret geometrically the action of l . When is l an *isometry*? In other words, when does l preserve distance? From

$$|l(w) - l(z)| = |a| |w - z|, \quad (2.3)$$

we see that l is an isometry if and only if $|a| = 1$. If $a = 1$, we term l the *translation through b* . We then note that if, in particular, $b \neq 0$, the line $L(c, b)$ is mapped onto itself by l . If $|a| = 1$ but $a \neq 1$, we term l a *rotation*.

When l is a rotation and $c = ac + b$, then

$$l(z) - c = a(z - c). \quad (2.4)$$

Clearly l preserves distance from c .

If $a > 0$ and $a \neq 1$, from (2.4) taken in the present context, we see that rays associated with (c, d) , $d \in \mathbb{C} - \{0\}$, are mapped onto themselves by l . Again from (2.4) we see that distances from c are multiplied by a . A linear map l for which $a > 0$ and there exists a *fixed point*, that is, a point c such that $l(c) = c$, is called a *homothety*. It is immediate that a linear map of $\mathbb{C}$ into $\mathbb{C}$ which is not constant is a translation, a rotation, a homothety, or the composition of a homothety and a rotation. It suffices to consider the case where $a \in \mathbb{C} - \mathbb{R}^+$. Let c denote the fixed point of l . Let $m(z) = c + (a/|a|)(z - c)$; let $n(z) = c + |a|(z - c)$. We have $l = n \circ m$. The converse statement is immediate.

EXERCISES

2.8. The linear maps of $\mathbb{C}$ into $\mathbb{C}$ with coefficient $a \neq 0$ map $\mathbb{C}$ univalently onto itself and constitute a group when taken together with composition. Show that the following are subgroups: the set of translations, the set of homotheties having a given fixed point, the set of rotations having a given fixed point together with the identity. What about the set of all homotheties, or all rotations together with the identity map?

2.9. When $f: X \rightarrow X$, we define f_n , the n th iterate of f , by $f_0 = \iota$, the identity map, $f_{n+1} = f \circ f_n$, $n \in \mathbb{N}$. Show that in the case of l , $l_n(z) = z + nb$ if $a = 1$, while if $a \neq 1$, $l_n(z) = c + a^n(z - c)$ where c is the fixed point of l .

3. LIMITS

We are concerned with sequences $(a_n): \mathbb{N} \rightarrow \mathbb{C}$. The definitions for the notions of a *limit* and of a *Cauchy sequence* are mere paraphrases of those given in elementary real analysis with absolute value now being taken in the sense of $\mathbb{C}$. With this agreed, given (a_n) and $\alpha \in \mathbb{C}$, the statement “ (a_n) has limit α ” needs no gloss nor do the notations “ $\lim_{n \rightarrow \infty} a_n$,” “ $\lim a_n$.” We say that (a_n) *possesses limit* ∞ , $\lim_{n \rightarrow \infty} a_n = \infty$, provided that $\lim_{n \rightarrow \infty} |a_n| = +\infty$. A set in $\mathbb{C}$ is termed *bounded* provided that it is contained in $\Delta(0; r)$ for some positive r . A function, in particular a sequence, with image in $\mathbb{C}$ is termed *bounded* provided that its image is. A sequence (a_n) is said to have a *finite limit* provided that it has limit α for some $\alpha \in \mathbb{C}$.

Certain routine verifications are readily carried out: (1) a sequence has at most one finite limit; (2) the usual rules for sum, difference, product, and quotient hold; (3) if a sequence has limit $\alpha \in \mathbb{C}$ (respectively ∞), then so does

every subsequence; (4) a sequence possessing a finite limit is bounded; a sequence possessing limit ∞ is unbounded; (5) for real sequences the notions of finite limit and Cauchy sequence agree with those given in real analysis; (6) given (a_n) and $\alpha \in \mathbb{C}$, $\lim a_n = \alpha$ if and only if $\lim \operatorname{Re} a_n = \operatorname{Re} \alpha$ and $\lim \operatorname{Im} a_n = \operatorname{Im} \alpha$; (7) (a_n) possesses a finite limit if and only if (a_n) is a Cauchy sequence.

It is well to point out a geometric interpretation of a limit: $\lim a_n = \alpha \in \mathbb{C}$ if and only if for each positive η it is the case that

$$\{a_{m+n}: n \in \mathbb{N}\} \subset \Delta(\alpha; \eta)$$

for some $m \in \mathbb{N}$, or equivalently that

$$\{n: a_n \notin \Delta(\alpha; \eta)\}$$

is bounded. We note further that if $\lim a_n = \alpha (\in \mathbb{C})$, then $(\varepsilon(n)) \downarrow$, $\lim \varepsilon(n) = 0$, where $\varepsilon(n) = \sup\{|a_{n+k} - \alpha|: k \in \mathbb{N}\}$.

EXERCISES

3.1. Check assertions (1)–(7). A link between the real and complex situations is furnished by the inequality $|\operatorname{Re} a|, |\operatorname{Im} a| \leq |a| \leq |\operatorname{Re} a| + |\operatorname{Im} a|$. This observation is useful for (6) and (7).

3.2. Given $\lim a_n = 0$. Let $b_n = (\sum_0^n a_k)/(n+1)$. Show that $\lim b_n = 0$. Hint: For fixed $m \in \mathbb{N}$ write

$$b_n = \frac{1}{n+1} \sum_0^m a_k + \frac{1}{n+1} \sum_{m+1}^n a_k, \quad n \geq m+1,$$

and estimate the size of b_n .

3.3. Given $\lim a_n = \alpha \in \mathbb{C}$. Let $c_n = a_n - \alpha$ so that $\lim c_n = 0$. Taking the notation of Ex. 3.2. in the present context show that $\lim b_n = \alpha$. This result may be stated: if (a_n) possesses a finite limit, then the sequence whose n th term is the arithmetic mean of $a_0, \dots, a_n$ possesses the same limit (Cesàro). Show by the example of $((-1)^n)_{n \in \mathbb{N}}$ that there exist sequences (a_n) not possessing a limit for which the corresponding (b_n) do possess finite limits.

Sequences of functions. It should be emphasized that when we are dealing with limit problems for sequences of functions, the nature of the common domain of the functions in question is not important initially. It is only when we are interested in whether or not such properties as continuity are preserved under limit operations that the nature of the domain enters our considerations; obviously we need a medium where the property in question makes sense.

Let there be given a set X and functions $f, f_n, n \in \mathbb{N}$, mapping X into $\mathbb{C}$. We say that (f_n) tends pointwise to f or that f is the pointwise limit of (f_n) provided

that $\lim f_n(x) = f(x)$ for each $x \in X$. We say that (f_n) *tends uniformly to f* or that f is the *uniform limit of (f_n)* provided that for each positive η there exists $m \in \mathbb{N}$ such that

$$|f(x) - f_n(x)| < \eta \quad (3.1)$$

for $x \in X, n \geq m$. We see that if f is the uniform limit of (f_n) and $X \neq \emptyset$, then for n sufficiently large the set

$$\{|f(x) - f_{n+k}(x)| : x \in X, k \in \mathbb{N}\}$$

is bounded, $\neq \emptyset$. Let $\varepsilon(n)$ denote the supremum of this set when n is large. For some $m \in \mathbb{N}$, $\lim_{n \rightarrow \infty} \varepsilon(m+n) = 0$. The converse is immediate.

Given $Y \subset X$, we say that f_n *tends pointwise* (resp., *uniformly*) *to f on Y* provided that $f_n|Y$ tends pointwise (resp., uniformly) to $f|Y$.

EXERCISES

3.4. If (f_n) tends uniformly to f , then (f_n) tends to f pointwise. Show by example that the converse is false.

3.5. Suppose that f_n tends uniformly to f and g_n tends uniformly to g , g and g_n mapping X into $\mathbb{C}$. Show that $(f_n + g_n)$ tends uniformly to $f + g$ but that $(f_n g_n)$ need not tend uniformly to a complex-valued function with domain X .

It is to be observed that the following definitions are in force. Let A denote a ring, let $f, g: X \rightarrow A$. Then by $f + g$ is meant the map taking x into $f(x) + g(x)$ and by fg is meant the map taking x into $f(x)g(x)$. With addition and multiplication so defined the set of $f: X \rightarrow A$ is a ring, commutative when A is. Given $\alpha \in A$, we define αf as the map taking x into $\alpha f(x)$. [By license the αf notation will apply when $\alpha \in \mathbb{C} - \{0\}$ and A is replaced by $\hat{\mathbb{C}}$, the extended complex plane (§6, Ch. III), αf being given as in the preceding sentence by the pointwise product, or when $\alpha \in \mathbb{C}$ and f is meromorphic (§4, Ch. VI) on an open set O , αf in this case being construed as the product in the sense of meromorphic functions of the constant with domain O taking the value α and the function f . Also, here and elsewhere the obvious conventions prevail when $X = \emptyset$; to be sure, this situation is not of great concern to us.]

To show failure for the product let $X =]0, 1]$, $f_n(x) = g_n(x) = x^{-1} + (n+1)^{-1}$.

Show that if in addition f and g are bounded, then $f_n g_n$ tends uniformly to fg . Show that if $|f_n(x)|, |f(x)| \geq m > 0$, then (h_n) tends uniformly to h where $h_n(x) = [f_n(x)]^{-1}$, $h(x) = [f(x)]^{-1}$.

Cauchy sequences. Given $f_n: X \rightarrow \mathbb{C}$, $n \in \mathbb{N}$. We say that (f_n) is a *pointwise Cauchy sequence* provided that $(f_n(x))$ is a Cauchy sequence for each $x \in X$. Such a sequence possesses a unique pointwise limit. We say that (f_n) is a *uniform Cauchy sequence* provided that for each positive η there exists $m \in \mathbb{N}$ such that

$$|f_p(x) - f_q(x)| < \eta \quad (3.2)$$

for $x \in X$, p and $q \geq m$. We see that if (f_n) is a uniform Cauchy sequence and $X \neq \emptyset$, then for n large the set

$$\{|f_p(x) - f_q(x)|: x \in X, p, q \geq n\}$$

is bounded, $\neq \emptyset$. Let $\omega(n)$ denote its supremum when n is large. For some $m \in \mathbf{N}$, $\lim_{n \rightarrow \infty} \omega(m+n) = 0$. The converse is immediate.

Given $Y \subset X$, we say that (f_n) is *pointwise* (resp., *uniformly*) *Cauchy on Y* provided that $(f_n|Y)$ is pointwise (resp., uniformly) Cauchy.

It is clear that a uniform Cauchy sequence is a pointwise Cauchy sequence and hence that it possesses a unique pointwise limit, say f . There can be no other candidate for the uniform limit.

EXERCISE

3.6. Show that f is in fact the uniform limit.

4. INFINITE SERIES

An infinite series is a sequence associated with a given sequence of a certain kind in a special way that we specify presently. Indeed, given $(a_n): \mathbf{N} \rightarrow G$ where G is an abelian group (in the first applications G is $\mathbf{C}$ considered additively or the set of complex-valued functions with a given domain X), by the *infinite series associated with (a_n)* is meant the sequence (s_n) where

$$s_n = \sum_0^n a_k.$$

We denote (s_n) by $\sum_0^\infty a_k$. s_n is termed the *n th partial sum of $\sum_0^\infty a_k$* .

For the sake of convenience we define $\sum_q^\infty a_k$ as $\sum_0^\infty a_{q+k}$ when $q \in \mathbf{Z}$ and (a_k) is a sequence mapping $\{k \in \mathbf{Z}, k \geq q\}$ into G and refer other subsequent use (to denote sum) of the notation $\sum_q^\infty a_k$ to the corresponding use of $\sum_0^\infty a_{q+k}$. Of course, we can equally well define $\sum_q^\infty a_k$ as $(s_n)_{n \geq q}$, where $s_n = \sum_q^n a_k$, in the first instance, and for subsequent limit questions refer directly to sequences with domain $\{k \in \mathbf{Z}, k \geq q\}$ using the appropriate definition.

To continue, we shall want to be more specific about G . We consider the numerical case ($G = \mathbf{C}$) first. We say that $\sum_0^\infty a_k$ is *convergent* provided that (s_n) possesses a finite limit, and in this case we denote $\lim_{n \rightarrow \infty} s_n$ by $\sum_0^\infty a_k$ —a concession to tradition since, in spite of the double use of the notation, there is no danger of confusion when the notation is taken in context. We term

$\lim_{n \rightarrow \infty} s_n$ the sum of $\sum_0^\infty a_k$ (now a sequence). We say that $\sum_0^\infty a_k$ is *divergent* provided that (s_n) does not possess a finite limit. We say that $\sum_0^\infty a_k$ *converges absolutely* provided that $\sum_0^\infty |a_k|$ is convergent.†

We next consider the case where the a_n are functions mapping a given set X into $\mathbb{C}$. We introduce $\{x: \sum_0^\infty a_k(x) \text{ converges}\}$. This set is termed the *convergence set* of $\sum_0^\infty a_k$. If the convergence set is X itself, then, trivially, $\sum_0^\infty a_k$ possesses a pointwise limit. In this case we say that $\sum_0^\infty a_k$ *converges pointwise* (on X) and we term the pointwise limit in question the (*pointwise*) *sum* of $\sum_0^\infty a_k$ and denote it by $\sum_0^\infty a_k$. We say that $\sum_0^\infty a_k$ *converges absolutely* (on X) provided that $\sum_0^\infty |a_k|$ converges pointwise.‡ We say that $\sum_0^\infty a_k$ *converges uniformly* provided that it possesses a uniform limit. The limit in question is called the *uniform sum* of $\sum_0^\infty a_k$. Given $Y \subset X$, we say that $\sum_0^\infty a_k$ *converges* (*converges uniformly*) *on* Y provided that $\sum_0^\infty (a_k|Y)$ converges (*converges uniformly*). In the respective cases, the limit (uniform limit) in question is called the *sum* (*uniform sum*) of $\sum_0^\infty a_k$ on Y . *Absolute convergence on* Y is correspondingly defined with “absolutely” replacing “uniformly.”

K. Knopp's *Infinite Series* [68] is to be recommended as an excellent account of the classical theory of infinite series and power series.

The following exercises serve for reviewing and for correlating the present material and the reader's earlier experience with infinite series.

EXERCISES

In Exs. 4.1–4.5 it is understood that $a_k \in \mathbb{C}$.

4.1. If $\sum_0^\infty a_k$ is convergent, then $\lim a_k = 0$.

4.2. If $a_k \geq 0$, $k \in \mathbb{N}$, then either $\sum_0^\infty a_k$ is convergent or $\lim s_k = +\infty$.

4.3. If $0 \leq b_k \leq a_k$, $k \in \mathbb{N}$, and $\sum_0^\infty a_k$ converges, then $\sum_0^\infty b_k$ converges and $\sum_0^\infty b_k \leq \sum_0^\infty a_k$.

4.4. If $\sum_0^\infty a_k$ converges absolutely, then $\sum_0^\infty a_k$ converges. Prove this theorem in the following two ways: (a) by showing that (s_n) is a Cauchy sequence, (b) by considering the following series:

$$\sum_0^\infty (\operatorname{Re} a_n)^+, \quad \sum_0^\infty (\operatorname{Re} a_n)^-, \quad \sum_0^\infty (\operatorname{Im} a_n)^+, \quad \sum_0^\infty (\operatorname{Im} a_n)^-.$$

4.5. Given $\sum_0^\infty a_k$ absolutely convergent, $\sum_0^\infty b_k$ divergent, b_k positive, $k \in \mathbb{N}$. Show that $\liminf_{n \rightarrow \infty} |a_n/b_n| = 0$.

4.6. Given $k \in \mathbb{N}$, let $f_k(z) = z^k$, $z \in \mathbb{C}$. Determine the convergence set of $\sum_0^\infty f_k$ (geometric series). Hint: Recall the identity $(\sum_0^n z^k)(1 - z) = 1 - z^{n+1}$.

4.7. Given $a_k > 0$, $k \in \mathbb{N}$. If $\limsup_{k \rightarrow \infty} a_{k+1}/a_k < 1$, then $\sum_0^\infty a_k$ is convergent. If

† Cautionary note: In spite of the sound of it, the fact that a series converges when it converges absolutely requires proof!

‡ We denote by $|f|$ the composition of the absolute value function and $f: X \rightarrow \mathbb{C}$.

$\liminf_{k \rightarrow \infty} a_{k+1}/a_k > 1$ (or $\lim a_{k+1}/a_k = +\infty$), then $\sum_0^\infty a_k$ is divergent. Here $\liminf a_{k+1}/a_k$ is construed to be in $\mathbf{R}$.

4.8. The Weierstrass M -test. Given $M_k \geq 0$, $f_k: X \rightarrow \mathbf{C}$, $k \in \mathbf{N}$. Suppose that (a) $\sum_0^\infty M_k$ is convergent, (b) $|f_k(x)| \leq M_k$, $x \in X$, $k \in \mathbf{N}$. Then $\sum_0^\infty f_k$ converges absolutely and converges uniformly. Hint: For uniformity show that $\sum_0^\infty f_k$ is uniform Cauchy. [This is the celebrated criterion for absolute and uniform convergence due to Weierstrass. “ M ” for “majorant.” The M -test is very useful in practice.]

4.9. Given $0 < r < 1$. Show that $\sum_0^\infty f_k$ of Ex. 4.6 converges absolutely and uniformly on $\{|z| \leq r\}$.

5. POWER SERIES

This section is our first serious encounter with complex function theory. We are concerned with a special instance of *infinite series of functions*. We are given $(a_n): \mathbf{N} \rightarrow \mathbf{C}$. We let $X = \mathbf{C}$ and introduce $f_n(z) = a_n z^n$, $z \in \mathbf{C}$. The series $\sum_0^\infty f_k$ is termed the *power series* with k th *coefficient* a_k . We denote both this power series and its sum at a point z of its convergence set by $\sum_0^\infty a_k z^k$.† Power series play an important role in complex function theory, their long history reaching back to the eighteenth century. They furnish methods of investigation of great value in the theory of differential equations, as we shall see later.

It should be kept in mind that in the study of power series the special case $a_n = 1$ (geometric series) is very important in that it is an instrument of comparison about which we have very complete information.

Let us turn to the question of the nature of the *convergence set* of a power series. It is easy to construct power series with convergence set $\{0\}$ as well as power series with convergence set $\mathbf{C}$. It suffices to take $a_n = n!$ for a power series of the first kind and $a_n = (n!)^{-1}$ for a power series of the second kind.‡

Suppose now that we are concerned with a power series $\sum_0^\infty a_k z^k$ for which neither of these extreme situations arises. There exist $c, d \in \mathbf{C} - \{0\}$ such that $\sum_0^\infty a_k c^k$ converges and $\sum_0^\infty a_k d^k$ diverges. As a first step in our study we shall prove the following lemma.

Lemma 5.1: *Let $0 < r < |c|$. Then $\sum_0^\infty a_k z^k$ converges absolutely and converges uniformly on $\{|z| \leq r\}$.*

PROOF: $(a_k c^k)$ is bounded. Let M be a positive number such that $|a_k c^k| \leq M$, $k \in \mathbf{N}$. Then for $|z| \leq r$ we have

$$|a_k z^k| = \left| a_k c^k \left(\frac{z}{c} \right)^k \right| \leq M \left(\frac{r}{|c|} \right)^k, \quad k \in \mathbf{N}. \quad (5.1)$$

† Given $z \in \mathbf{C}$, of course $\sum_0^\infty a_k z^k$ denotes as well the infinite series associated with $(a_n z^n)$. The meaning of the notation should be clear in any given case from the context.

‡ The utterly trivial $a_n = 0$ would do for a series of the second kind, but we wish to point out a nontrivial instance.

The lemma follows by virtue of the Weierstrass M -test and the convergence properties of the geometric series.

Now let $E = \{ |z| : \sum_0^\infty a_k z^k \text{ converges} \}$. We note that $|c| \in E$. Further, $|d|$ is an upper bound of E , for by the lemma if $|z| > |d|$, $\sum_0^\infty a_k z^k$ diverges. Hence E possesses a positive supremum $\rho \geq |c|$. If $|z| < \rho$, then there exists w , $|z| < |w| \leq \rho$, such that $\sum_0^\infty a_k w^k$ converges. Hence in the present case we see that the convergence set of $\sum_0^\infty a_k z^k$ contains $\Delta(0; \rho)$ and is contained in $\{ |z| \leq \rho \}$.

We are led to the following theorem.

Theorem 5.1: *The convergence set of the power series $\sum_0^\infty a_k z^k$ satisfies one of the following mutually exclusive conditions: (a) it is $\{0\}$, (b) it is $\mathbf{C}$, (c) there exists positive ρ (necessarily unique) such that $\Delta(0; \rho)$ is contained in the convergence set and $\{ |z| \leq \rho \}$ contains it.*

In case (b) $\sum_0^\infty a_k z^k$ converges absolutely and converges uniformly on $\{ |z| \leq r \}$ for each positive r ; in case (c) this is true for r , $0 < r < \rho$.

EXERCISE

5.1. Furnish the details of the proof which are not contained in the preceding discussion.

According as (a), (b), (c) holds, we say that the *radius of convergence* of $\sum_0^\infty a_k z^k$ is 0, $+\infty$, ρ . In case (c) we term $C(0; \rho)$ the *circumference of convergence*. The geometric series has radius of convergence 1. Its convergence set is $\Delta(0; 1)$. It is easy to give examples of power series with radius of convergence 1 which converge on $C(0; 1)$, in fact, even uniformly on $\{ |z| \leq 1 \}$. It suffices to take $a_n = (n+1)^{-2}$. If we take $a_n = (-1)^n (n+1)^{-1}$, we have a power series with radius of convergence 1 which converges for $z = 1$ and diverges for $z = -1$. In general, the study in the third case of the points of $C(0; \rho)$ at which $\sum_0^\infty a_k z^k$ converges and of the nature of the sum function is delicate and complicated.† The sum is, however, well behaved in $\Delta(0; \rho)$ for case (c) and in $\mathbf{C}$ for case (b) as we shall see shortly. Before we turn to this question, we shall continue our study of the radius of convergence by showing how it may be determined in terms of (a_n) . The solution is due to Cauchy and Hadamard. Into it two themes enter: the geometric series and the notion of limit superior.

Suppose that $\sum_0^\infty a_k z^k$ has radius of convergence 0. Given r positive,

† For some classical examples, cf. [70].

$\{k: |a_{k+1}|r^{k+1} > 1\}$ is unbounded. Otherwise for some $m \in \mathbb{N}$ we should have

$$|a_{k+1}z^{k+1}| = \left| a_{k+1}r^{k+1} \left(\frac{z}{r} \right)^{k+1} \right| \leq \left(\frac{|z|}{r} \right)^{k+1},$$

$k \geq m$, and the radius of convergence of $\sum_0^\infty a_k z^k$ would not be 0. Hence $|a_{k+1}|r^{k+1} > 1$, and, consequently, $\sqrt[k+1]{|a_{k+1}|} > r^{-1}$ holds for an unbounded set of k and hence

$$\sup\{\sqrt[k+1]{|a_{k+1}|}: k \in \mathbb{N}\} = +\infty.$$

That is, $\limsup_{k \rightarrow \infty} \sqrt[k+1]{|a_{k+1}|} = +\infty$.

Suppose that $\sum_0^\infty a_k z^k$ has radius of convergence $+\infty$. Given r positive, since $\sum_0^\infty a_k r^k$ is convergent, there exists $m \in \mathbb{N}$ such that $|a_{k+1}|r^{k+1} \leq 1$ and consequently $\sqrt[k+1]{|a_{k+1}|} \leq r^{-1}$ for $k \geq m$. Hence, since r is an arbitrary positive number, $\lim_{k \rightarrow \infty} \sqrt[k+1]{|a_{k+1}|} = 0$.

Suppose that $\sum_0^\infty a_k z^k$ has positive radius of convergence ρ . Let $0 < s < \rho < t$. Then $\{k: |a_{k+1}|s^{k+1} > 1\}$ is bounded since $\sum_0^\infty a_k s^k$ is convergent, and $\{k: |a_{k+1}|t^{k+1} > 1\}$ is unbounded as we see on appeal to the reasoning used in the case of radius of convergence 0—here we should be led to the convergence of $\sum_0^\infty a_k z^k$ on $\Delta(0; t)$. From the first assertion we see that $(\sqrt[k+1]{|a_{k+1}|})$ is bounded. Suppose that $\limsup_{k \rightarrow \infty} \sqrt[k+1]{|a_{k+1}|} = \alpha$. From the first assertion $\alpha \leq s^{-1}$ and from the second $\alpha \geq t^{-1}$. Given the arbitrariness of s and t , we see that

$$\limsup_{k \rightarrow \infty} \sqrt[k+1]{|a_{k+1}|} = \rho^{-1}.$$

It is now easy to see by the disjunction of the above conclusions that the following theorem holds.

Theorem 5.2 (Cauchy–Hadamard): *If $\lim_{k \rightarrow \infty} \sqrt[k+1]{|a_{k+1}|} = 0$, then $\sum_0^\infty a_k z^k$ has radius of convergence $+\infty$. If $\limsup_{k \rightarrow \infty} \sqrt[k+1]{|a_{k+1}|} = +\infty$, then $\sum_0^\infty a_k z^k$ has radius of convergence 0. If $(\sqrt[k+1]{|a_{k+1}|})$ has limit superior α , where α is a positive number, then $\sum_0^\infty a_k z^k$ has radius of convergence α^{-1} .*

If we anticipate the language of §6, Ch. III, where we shall have the *extended* real line available, we may put Theorem 5.2 in the tidier form: the radius of convergence of $\sum_0^\infty a_k z^k$ is $1/(\limsup_{k \rightarrow \infty} \sqrt[k+1]{|a_{k+1}|})$.

EXERCISES

5.2. Let r be positive. If $\{k: \sqrt[k+1]{|a_{k+1}|} > r\}$ is unbounded, then either the radius of convergence is zero, or else it is positive but no greater than r^{-1} .

5.3. What is the radius of convergence when $a_k = k^k, k^2, k^{-2}, k^{-k}$? Let $b_k = 1$ when $k = m^2$ for some $m \in \mathbb{N}$ and otherwise let $b_k = 0$. What is the radius of convergence when $a_k = k^k b_k, k^2 b_k, k^{-2} b_k, k^{-k} b_k$?

5.4. The convergence sets of $\sum_0^\infty a_k z^k$ and $\sum_0^\infty a_{m+k} z^k, m \in \mathbb{N}$, are the same.

5.5. What is the radius of convergence when $a_k = \sin k$? [We presuppose for the sake of the exercise properties of the trigonometric functions which will be officially introduced in Ch. IV. The reader should check to see whether any properties of the sine function are invoked which do not come under the arithmetic treatment of Ch. IV. The irrationality of π is not to be assumed. Hint: Show that

$$\limsup_{k \rightarrow \infty} |\sin k| > 0$$

using the addition theorem for the sine and the fact that $\sin 1 \neq 0$.]

5.6. What can be said about the radii of convergence of $\sum_0^\infty (\operatorname{Re} a_k) z^k$ and $\sum_0^\infty (\operatorname{Im} a_k) z^k$ when the radius of convergence of $\sum_0^\infty a_k z^k$ is known?

5.7. The radii of convergence of $\sum_0^\infty a_k z^k$ and $\sum_0^\infty b_k z^k$ are known. Investigate the radius of convergence of $\sum_0^\infty c_k z^k$ for the following situations: $c_k = a_k + b_k, c_k = a_k b_k, c_k = a_k / b_k$. In the last case $b_k \neq 0, k \in \mathbb{N}$. Construct examples to test the precision of your statements.

The reader should recall from elementary real analysis the theorem concerning the continuity of the uniform limit of a sequence of continuous functions. Actually, this theorem is a special case of a general theorem considered in Ex. 5.3, Ch. III, which subsumes the case of complex-valued functions. For the present we note that the definition of continuity for complex-valued functions with domain in $\mathbb{C}$ is merely a paraphrase of that used in the real case and that the uniform limit property holds. Suppose that the radius of convergence of $\sum_0^\infty a_k z^k$ is not 0. We see by Theorem 5.1 that f , in case (b) the sum and in case (c) the sum on $\Delta(0; \rho)$, is *continuous*. Many other properties of f may be inferred.

We continue by noting that if $\{k: a_k \neq 0\} \neq \emptyset$, then there is a least member m . Now

$$f(z) = \sum_0^\infty a_k z^k = z^m g(z), \quad (5.2)$$

where g has the same context relative to $\sum_0^\infty a_{m+k} z^k$ as does f for $\sum_0^\infty a_k z^k$. By virtue of the fact that $g(0) \neq 0$ and g is continuous at 0, we conclude that there exists r positive such that $g(z) \neq 0$ for all $z \in \Delta(0; r)$.† Hence $f(z) \neq 0, 0 < |z| < r$. We conclude:

If $\inf\{|z|: f(z) = 0, z \neq 0\} = 0$, then $a_k = 0, k \in \mathbb{N}$.

The hypothesis is fulfilled, in particular, if $f(z) = 0, |z| < r$, for some positive

† Note that there exists r positive such that $|g(0) - g(z)| < |g(0)|$ for all $z \in \Delta(0; r)$. For such $z, g(z) \neq 0$.

r or all the more so if $f(z) = 0$ for all z in the domain of f . The converse is trivial.

There is an important corollary, the identity theorem for power series. Suppose that $\sum_0^\infty a_k z^k$ and $\sum_0^\infty b_k z^k$ converge, $|z| < r$, where r is positive, and have sums on $\Delta(0; r)$ which are equal. Then $a_k = b_k$, $k \in \mathbb{N}$. It suffices to consider $\sum_0^\infty (a_k - b_k)z^k$. It would, of course, be enough to require that

$$\inf \left\{ |z| : \sum_0^\infty a_k z^k = \sum_0^\infty b_k z^k, z \neq 0 \right\} = 0.$$

Note that the corollary yields the unique representation of a polynomial function in the form $\sum_0^\infty a_k z^k$. Here $\{k : a_k \neq 0\}$ is bounded.

We now give some exercises whose object will be to provide experience in reasoning with power series.

EXERCISES

5.8. Given $(a_k) : \mathbb{N} \rightarrow \mathbb{C}$ satisfying $a_1 = 1$, $\sum_2^\infty k |a_k| \leq 1$. Then $f(z) = \sum_0^\infty a_k z^k$, $|z| < 1$, is univalent. Hint: Let $z, w (\neq z) \in \Delta(0; 1)$, consider $f(w) - f(z)$ and represent it as a product $(w - z)A$ and obtain a lower estimate for $|A|$ which renders the fact that $A \neq 0$ clear.

Deduce the following corollary: Let $\sum_0^\infty a_k z^k$ have radius of convergence not zero and let $a_1 \neq 0$. There exists r positive such that the series has a univalent sum on $\Delta(0; r)$. Hint: Consider $(a_1 r)^{-1} \sum_0^\infty (a_k r^k) z^k$ for small positive r .

5.9. We consider $\sum_0^\infty a_k z^k$ converging for $|z| < r$, r positive, and denote the sum on $\Delta(0; r)$ by f . Let $u \in C(0; 1)$ and satisfy $u^{n+1} \neq 1$, $n \in \mathbb{N}$. Let $|z| < r$ and introduce $A_n(z) = (n+1)^{-1} \sum_0^n f(u^k z)$. Show that $\lim_{n \rightarrow \infty} A_n(z) = a_0 = f(0)$. Hint: It is convenient to represent A_n as the sum of a power series on $\Delta(0; r)$ and to control the m th partial sum and the difference of A_n and the m th partial sum.

Conclude that $|a_0| \leq M(s)$ where $0 < s < r$ and $|f(z)| \leq M(s)$, $|z| = s$. By considering $f(z)z^{-j}$, $j \in \mathbb{N}$, show that $|a_j|s^j \leq M(s)$, $j \in \mathbb{N}$. These are essentially the well-known *Cauchy inequalities*.

The indicated proof was given by A. Hurwitz [60] and uses implicitly an idea that occurs in the equipartition theorem (Gleichverteilungssatz) of H. Weyl [129; p. 111].

We need u with the stated property. This is a *sine qua non* of the proof. The reader with experience in countability arguments will see that the existence of u follows from the facts that $C(0; 1)$ is not countable and the set of z satisfying $z^{n+1} = 1$ for some $n \in \mathbb{N}$ is countable. Hurwitz [loc. cit.] exhibited such a u , namely $(2-i)/(2+i)$, showing that u has the stated property as follows. If $u^{n+1} = 1$ for some $n \in \mathbb{N}$, $(2-i)^{n+1} = (2+i)^{n+1} \in \mathbb{Z}$. Let

$$m = \min\{n+1 : (2+i)^{n+1} \in \mathbb{N}, n \in \mathbb{N}\}.$$

Then $(2+i)^m = 5^{m/2}$ and m is even. Consequently, $(2+i)^{m/2} = -5^{m/4}$ and m is divisible by 4. Consequently, $(2+i)^{m/4} = 5^{m/8}i$ or $-5^{m/8}i$ and m is divisible by 8. Note that $\operatorname{Re}(2+i)^n$ is odd for n even and even for n odd while $\operatorname{Im}(2+i)^n$ is even for n even and odd for n odd. Conclude that $\operatorname{Im}(2+i)^{m/4}$ is even. Contradiction.

5.10. Let $\sum_0^\infty a_k z^k$ converge on $\mathbb{C}$ and let f denote the sum. Suppose that there exist M positive, $\alpha \geq 0$, and $(r_n)^\dagger$, $0 < r_0$, $\lim r_n = +\infty$, such that $|f(z)| \leq M r_n^\alpha$, $|z| = r_n$, $n \in \mathbb{N}$. Show that $a_k = 0$, $k > \alpha$. We have as corollaries: (1) the theorem of Liouville that states: if $|f(z)| \leq M$, $z \in \mathbb{C}$, then f is constant; and (2) the result that if $\sum_0^\infty a_k z^k$ converges on $\mathbb{C}$ and $\{k: a_k \neq 0\}$ is unbounded, then the sum is not a quotient of polynomials. [Appeal is made to the notion of a real power of a positive number in the sense of real analysis; $a^b = \exp(b \log a)$, $a \in \mathbb{P}$, $b \in \mathbb{R}$.]

5.11. The Cauchy inequalities permit us to obtain many properties of power series and to study their behavior under limit processes. These questions will be explored in this and other exercises of this set.

Let r be positive. For each $n \in \mathbb{N}$ let f_n denote the sum on $\Delta(0; r)$ of a power series converging on $\Delta(0; r)$. Suppose that there exists $f: \Delta(0; r) \rightarrow \mathbb{C}$ such that for each s , $0 < s < r$, (f_n) tends uniformly on $\{|z| \leq s\}$ to f . Show that if $a_k^{(n)}$ is the k th coefficient of the power series with sum f_n on $\Delta(0; r)$, then $(a_k^{(n)})_{n \in \mathbb{N}}$ possesses a finite limit a_k , $k \in \mathbb{N}$, and $f(z) = \sum_0^\infty a_k z^k$, $|z| < r$. What is the situation when $\Delta(0; r)$ is replaced by $\mathbb{C}$?

5.12. Suppose that (a_k) and (b_k) are such that $\{k: a_k \neq 0\}$ and $\{k: b_k \neq 0\}$ are bounded. Show that

$$\left(\sum_0^\infty a_k z^k \right) \left(\sum_0^\infty b_k z^k \right) = \sum_0^\infty c_k z^k,$$

where $c_k = \sum_0^k a_j b_{k-j}$, $k \in \mathbb{N}$. Note that $\{k: c_k \neq 0\}$ is bounded.

5.13. Suppose now that $\sum_0^\infty a_k z^k$ and $\sum_0^\infty b_k z^k$ are power series subject only to the restriction that each converges on $\Delta(0; r)$, r positive. Let c_k be defined as in the previous exercise. Show that $\sum_0^\infty c_k z^k$ converges on $\Delta(0; r)$ and that

$$\sum_0^\infty c_k z^k = \left(\sum_0^\infty a_k z^k \right) \left(\sum_0^\infty b_k z^k \right)$$

(Cauchy multiplication). Hint: Apply Ex. 5.12 to $(\sum_0^n a_k z^k)(\sum_0^n b_k z^k)$ and thereupon use Ex. 5.11.

***5.14.** Given that $\sum_0^\infty a_k z^k$ converges on $\Delta(0; r)$ and has sum f on $\Delta(0; r)$ and that $\sum_0^\infty b_k z^k$ converges on $\Delta(0; s)$ and has sum g on $\Delta(0; s)$, r, s positive. Suppose further that $g[\Delta(0; s)] \subset \Delta(0; r)$. Show that $f[g(z)] = \sum_0^\infty c_k z^k$, $|z| < s$. The Weierstrass theorem concerning the maximum of a continuous real-valued function (cf. Theorem 2.2, Ch. III) is to be assumed. [c_k is not to be taken in the sense of the preceding exercise.]

Show that if $|g(z) - g(0)| < |g(0)|$, then $1/g(z) = \sum_0^\infty d_k z^k$, $|z| < s$, and show how the coefficients d_k may be determined. Hint: Take $f(z) = (1 - z)^{-1}$, $|z| < 1$, and consider $f[1 - g(z)/g(0)]$.

We now turn to two problems whose results yield information about the behavior of power series on the *circumference of convergence* and continuity properties of the sum. In both exercises the Abel identity is a fundamental tool.

EXERCISES

5.15. Given $(a_k)^\dagger$, $\lim a_k = 0$. Then the convergence set of $\sum_0^\infty a_k z^k$ contains $C(0; 1) - \{1\}$. In fact, $\sum_0^\infty a_k z^k$ converges uniformly on $\{|z| \leq 1, |z - 1| \geq r\}$,

$0 < r < 2$. Hint: Apply the Abel identity to $\sum_{m+1}^{\infty} a_k z^k$. A simple important example is given by $a_k = (k+1)^{-1}$.

What can be said when the assumption, $\lim a_k = 0$, is dropped?

5.16. (Abel's Theorem.) Given that $\sum_0^{\infty} a_k$ is convergent, $a_k \in \mathbb{C}$ (or equivalently that the convergence set of $\sum_0^{\infty} a_k z^k$ contains $\Delta(0; 1) \cup \{1\}$). Let $r \geq 1$ and let $E = \{|1-z| \leq r(1-|z|)\}$. We have $E \subset \Delta(0; 1) \cup \{1\}$. Show that f is continuous at 1, f being the sum of $\sum_0^{\infty} a_k z^k$ on E . Hint:

$$f(1) - f(z) = \sum_1^{\infty} a_k(1-z^k) = \sum_1^m a_k(1-z^k) + \sum_{m+1}^{\infty} a_k(1-z^k).$$

Apply the Abel identity to $\sum_{m+1}^{\infty} a_k(1-z^k)$ with the $(1-z^k)$ playing the role of the a 's of the identity. Sketch the set E for $r = 1$ and $r > 1$.

We now consider two theorems which yield as a conclusion the hypothesis of the Abel theorem, that is, the convergence of a series $\sum_0^{\infty} a_k$ with complex terms when certain information is available on the allied power series $\sum_0^{\infty} a_k z^k$ and its sum f on $[0, 1[$. We suppose that $\sum_0^{\infty} a_k z^k$ converges on $\Delta(0; 1)$ and that $f| [0, 1[$ possesses a finite limit α at 1. Does $\sum_0^{\infty} a_k$ converge? Not always. Example: $\sum_0^{\infty} (-1)^k z^k$. The theorem that we shall prove is due to Tauber. It says: *If the condition $\lim(ka_k) = 0$ is added, then $\sum_0^{\infty} a_k = \alpha$.* Far more sophisticated results are known such as the celebrated Tauberian theorem of Littlewood [cf. 70]. Their study would carry us outside of our program.

PROOF OF TAUBER'S THEOREM: Let $s_n = \sum_0^n a_k$. The proof is based upon a skillful choice of a sequence $(x_n) \uparrow$, $0 < x_n < 1$, $n \in \mathbb{N}$, $\lim x_n = 1$ which insures that $\lim[f(x_n) - s_n] = 0$. To that end, we note that if $0 < x < 1$, then

$$f(x) - s_n = \sum_0^n a_k(x^k - 1) + \sum_{n+1}^{\infty} a_k x^k,$$

so that

$$\begin{aligned} |f(x) - s_n| &\leq \sum_0^n |a_k| (1 - x^k) + \omega(n+1) \sum_{n+1}^{\infty} \frac{x^k}{k} \\ &\leq (n+1) \omega(0) (1-x) + \frac{\omega(n+1)}{n+1} (1-x)^{-1} \end{aligned} \quad (5.3)$$

where

$$\omega(n) = \sup\{(n+k) |a_{n+k}| : k \in \mathbb{N}\}.$$

We suppose that $\omega(n) > 0$, $n \in \mathbb{N}$. Otherwise the theorem would be obvious.

Our choice of (x_n) is motivated by the following elementary fact: If A and B are positive, then the minimum value of $Ax + Bx^{-1}$, $x > 0$, is equal to

$2\sqrt{AB}$ and is attained at $\sqrt{B/A}$ and only at $\sqrt{B/A}$. This assertion follows from the identity

$$Ax + Bx^{-1} = \left(\sqrt{Ax} - \sqrt{\frac{B}{x}} \right)^2 + 2\sqrt{AB}, \quad x > 0. \quad (5.4)$$

Returning to the proof we take

$$1 - x_n = \frac{1}{n+1} \sqrt{\frac{\omega(n+1)}{\omega(0)}}$$

and conclude that

$$|f(x_n) - s_n| \leq 2\sqrt{\omega(0) \omega(n+1)}.$$

The theorem follows.

We obtain the theorem of Fejér when we replace in the theorem of Tauber the condition $\lim(ka_k) = 0$ by the condition $\sum_1^\infty k|a_k|^2$ converges. This condition may appear special at a first glance. It does have an interpretation in terms of area which is developed in the geometric theory of functions [43]. The proof of the Fejér theorem proceeds along lines similar to those of the Tauber theorem but calls for the use of the important *Cauchy inequality* [50] (not to be confused with the Cauchy inequalities), which it will be opportune to treat here thanks to (5.4):

Given $a_k, b_k \in \mathbf{R}$, $k = 1, \dots, n$. We have

$$\left(\sum_1^n a_k b_k \right)^2 \leq \left(\sum_1^n a_k^2 \right) \left(\sum_1^n b_k^2 \right) \quad (5.5)$$

(*Cauchy inequality*). Equality holds in (5.5) if and only if there exists (λ, μ) , λ, μ real not both zero, such that $\lambda a_k + \mu b_k = 0$, $k = 1, \dots, n$.

PROOF: We put aside the trivial cases: $a_k = 0$, all k ; $b_k = 0$, all k . In the remaining case, at all events

$$\sum_1^n \left(a_k \sqrt{x} - \frac{b_k}{\sqrt{x}} \right)^2 \geq 0, \quad x > 0, \quad (5.6)$$

so that

$$\left(\sum_1^n a_k^2 \right) x + \left(\sum_1^n b_k^2 \right) x^{-1} \geq 2 \sum_1^n a_k b_k, \quad x > 0.$$

Hence taking the minimum value on the left we obtain

$$\sum_1^n a_k b_k \leq \left(\sum_1^n a_k^2 \sum_1^n b_k^2 \right)^{1/2}$$

and replacing b_k by $-b_k$, a corresponding inequality. The two so obtained inequalities taken together yield (5.5). Equality in (5.5) yields, when we retrace our steps, that (5.6) is an equality for some x or else (5.6) with b_k replaced by $-b_k$ is. The stated proportionality follows. Equality in (5.5), when the stated (λ, μ) exists, is readily verified.

The following corollary is easy to see: *Given $(a_k), (b_k): \mathbf{N} \rightarrow \mathbf{R}$ such that $\sum_0^\infty a_k^2$ and $\sum_0^\infty b_k^2$ converge. Then $\sum_0^\infty a_k b_k$ converges absolutely and*

$$\left(\sum_0^\infty a_k b_k \right)^2 \leq \left(\sum_0^\infty a_k^2 \right) \left(\sum_0^\infty b_k^2 \right). \quad (5.7)$$

Equality holds in (5.7) if and only if there exists (λ, μ) as above such that $\lambda a_k + \mu b_k = 0, k \in \mathbf{N}$.

EXERCISES

5.17. Prove Fejér's theorem. Hint: Introduce $r(n) = [\sum_n^\infty k |a_k|^2]^{1/2}$. We put aside the trivial case $r(n) = 0, n$ large. With $0 < x < 1$, show, using the Cauchy inequality and its corollary, that

$$|f(x) - s_n| \leq n r(0)(1-x) + \frac{r(n+1)}{\sqrt{n+1}} (1-x)^{-1/2}, \quad n \in \mathbf{N}.$$

Show that a sequence (x_n) may be chosen advantageously (as in the proof of the Tauber theorem) by taking

$$1 - x_n = \frac{1}{n} \left[\frac{r(n+1)}{r(0)} \right]^{2/3}$$

when $n \geq 1$.

5.18. Construct (a_k) for which the hypotheses of the Tauber theorem but not of the Fejér theorem are fulfilled and vice versa.